

A Cross-Disciplinary Study Evaluating the Effectiveness of GenAI-Created Multiple Choice Questions

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| Can AI Help Create Assessments?

POTENTIAL BENEFITS

the promise

- + Faster question creation
- + Larger question banks
- + Reduced instructor workload

MAJOR CONCERNS

the risk

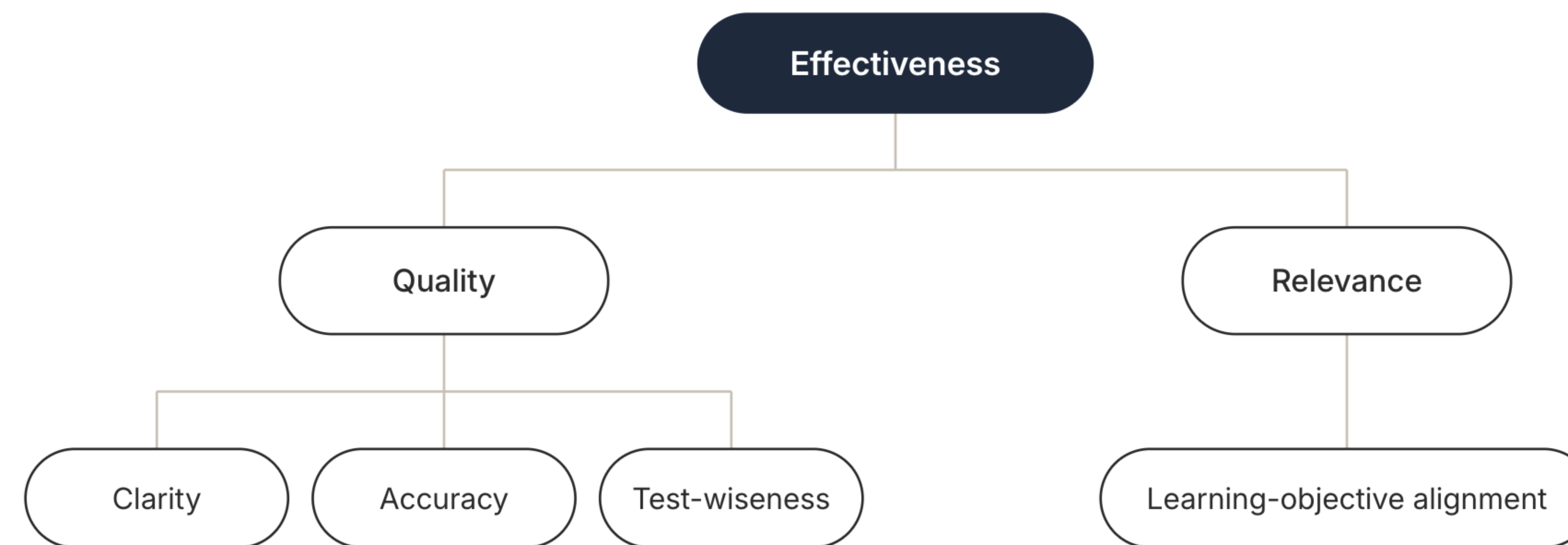
- Accuracy
- Question quality
- Learning objective alignment

RESEARCH QUESTION

Can GenAI reliably generate classroom-ready MCQs when guided by explicit pedagogical guardrails?

Classroom-ready = high quality + aligned to learning objectives

| How We Evaluated Effectiveness



A question is classroom-ready when it passes **both** quality and relevance.

Generation Was Guided by a Carefully Designed Prompt

Quality guardrails

WELL-BUILT

- ✓ One concept per question
- ✓ Simple, plain language
- ✓ Technically correct answers
- ✓ Parallel answer choices
- ✓ Avoid absolute terms

Relevance guardrails

ON-TARGET

- ✓ Focus on important concepts
- ✓ Emphasize core understanding
- ✓ Avoid trivial edge cases

Prompt design was intentionally aligned with **assessment best practices**.

| Study Overview

1,252

AI-GENERATED MCQS EVALUATED

60
SECTIONS

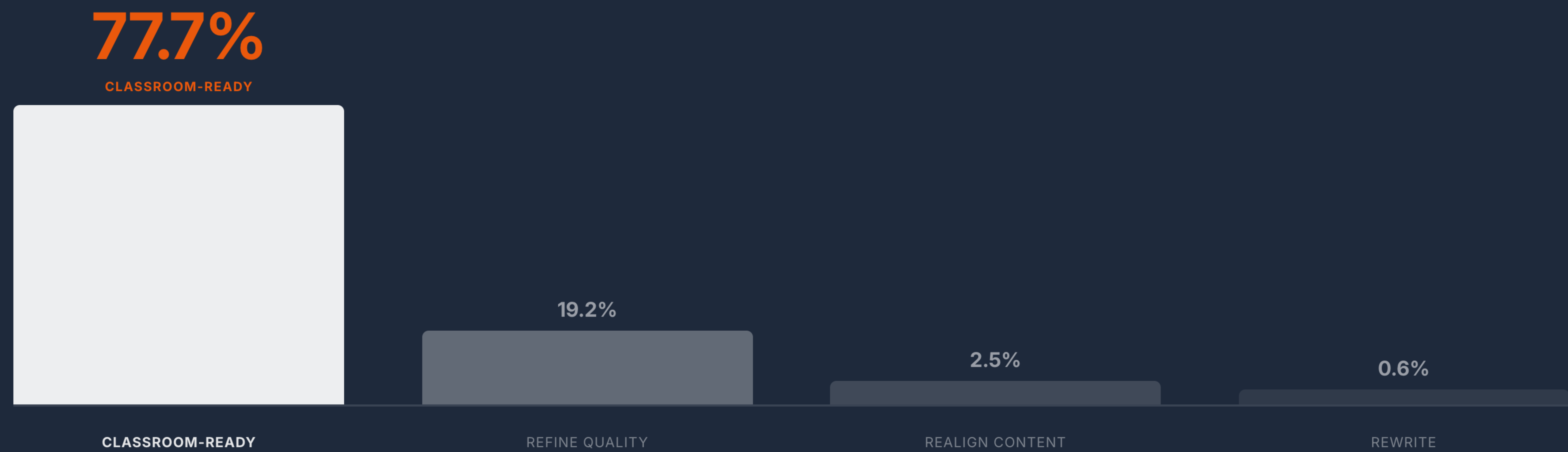
4
TEXTBOOKS

3
DISCIPLINES

Computer Science · Data Science · Mathematics

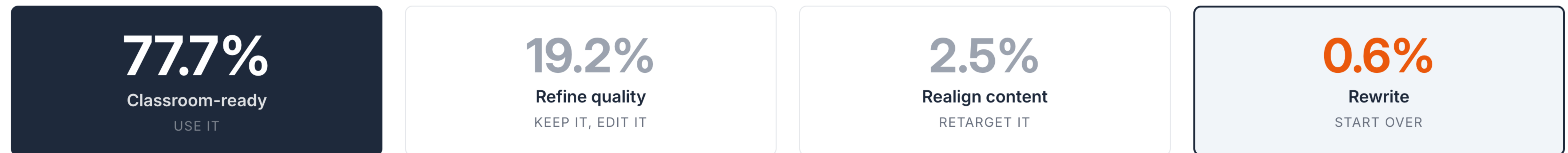
Each question was evaluated using our effectiveness framework across
quality and learning-objective alignment.

| The Headline Finding



n = 1,252 questions across three disciplines

Most Non-Passing Questions Needed Refinement, Not Replacement



Only 0.6% required starting over

What a Classroom-Ready Question Looks Like

In the expression $y = 3 + 2 * x$, which operation is performed first?

A. Addition of $3 + 2$

B. Multiplication of $2 * x$

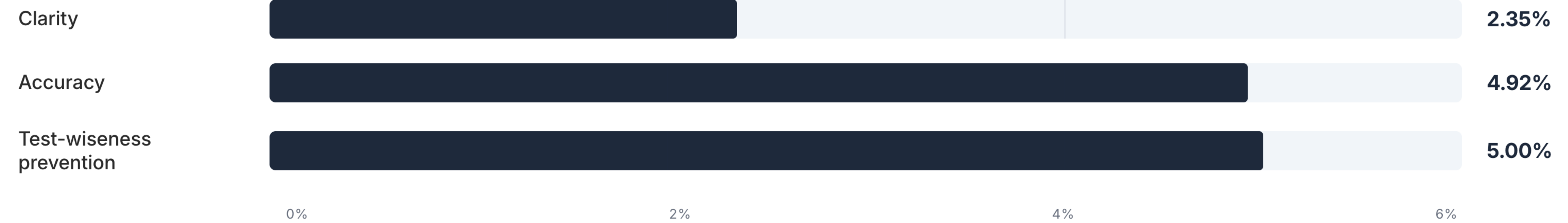
C. Assignment to y

D. Evaluation from right to left

Answer: **B**

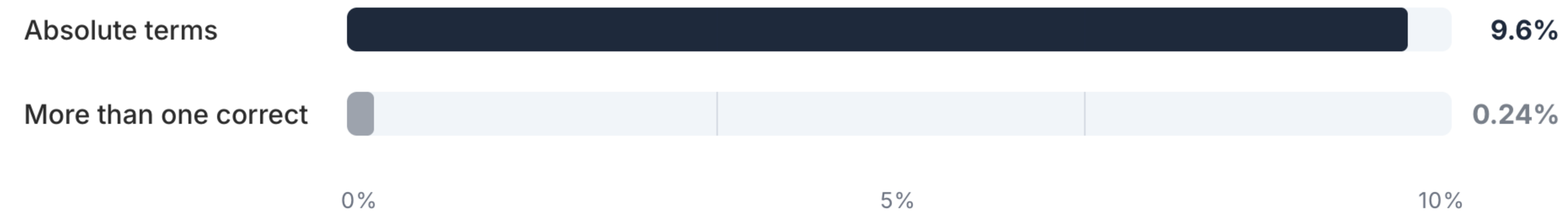
- ✓ Clear
- ✓ Accurate
- ✓ Learning-objective aligned
- ✓ No item-writing flaws

Where Did Failures Occur?



Failure rate per criterion check · AI-generated questions only · n = 1,230

Inside Accuracy: Almost All Overconfident Wording



Absolute terms account for **97.5%** of all accuracy failures.

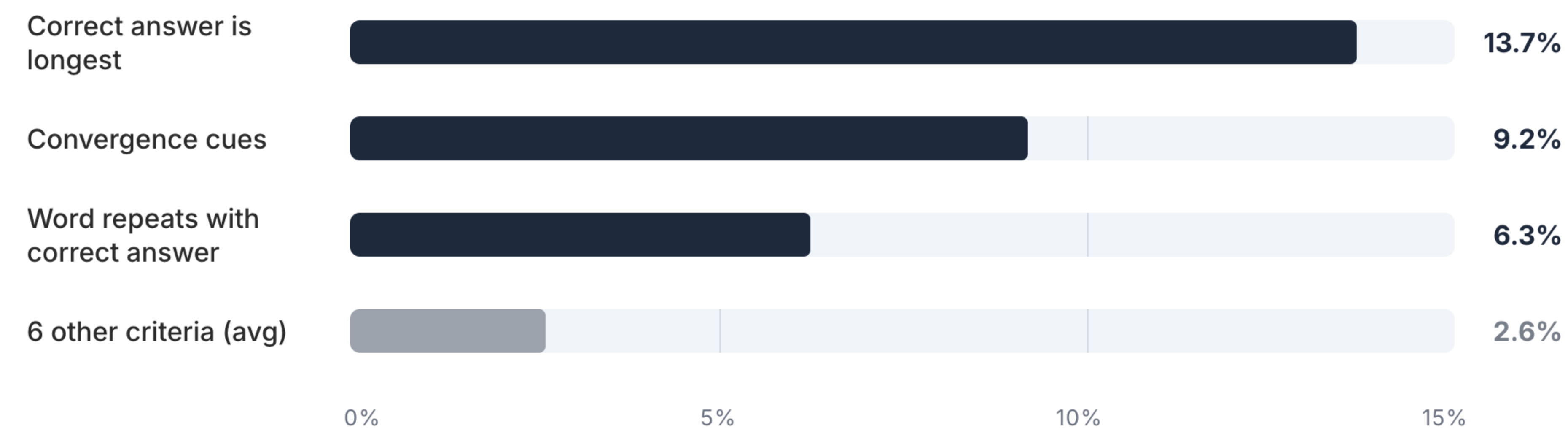
EXAMPLE: ABSOLUTE TERMS

Why must the left side of an assignment be a variable in programming?

- A. Because **only** variables can store values
- B. Because expressions always return false
- C. Because variables can store multiple values simultaneously
- D. Because expressions are only used on the right side

Why it fails: "Only" overstates the case — constants, object fields, and array elements also hold values.

Inside Test-Wiseness: Three Recurring Patterns



Top 3 patterns account for **65%** of all test-wiseness failures.

EXAMPLE: CORRECT ANSWER IS LONGEST

How can a programmer prevent an infinite loop?

- A. Use a loop with no body
- B. Ensure the loop variable or condition changes to eventually become False**
- C. Avoid using while loops
- D. Always start the loop condition as False

Why it fails: *The correct answer is visibly longer than every distractor — a tell that students learn to follow.*

| How Instructors Can Use This



- AI reduces blank-page work
- Human review remains important
- Review should focus on technical precision and wording
- The strongest use case is draft generation, not full automation

| Key Takeaways

- 01 Prompt guardrails matter.
- 02 **77.7%** of generated questions were classroom-ready.
- 03 Only **0.6%** required complete replacement.
- 04 AI is most useful as a draft-generation partner for instructors.

IN ONE SENTENCE

GenAI can be a practical assessment-design partner when guided by explicit pedagogical guardrails.

DISCUSSION

Questions?

77.7%

CLASSROOM-READY

*"Prompt
guardrails
matter."*

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